

MAT 180 - Midterm Project Write-Up

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Problem statement and motivation

Many college students have experienced the struggle of leaving a crowded classroom and trying to make it to their next class on time. Especially for classes and lecture halls that house large amounts of students at a time, trying to bypass people and make it through the tiny doors can often be difficult and even dangerous for themselves and other fellow students.

Thus, the goal of our project would be to analyze the congestion patterns of classrooms during end of class times to determine the best possible actions students can take to maximize the speed at which students can leave the classroom safely and efficiently.

Simplifying Assumptions

We introduce the following modeling assumptions in order to make this problem feasible to be modeled using simple mathematical concepts covered in class:

- **Like-minded students:** All students have the same behaviors (walking speed and personal space boundaries), and everyone wants to leave the classroom at the exact same time.
- **Greedy Students:** We assume the worst case scenario where all students prioritize minimizing their own evacuation time and act competitively. They maximize their walk speed v_{max} to push towards the door ahead of others, disregarding group efficiency.
- **Bounded 2D Continuous Domain:** The classroom is a flat, continuous square space with walls on all four sides
- **Point Exit with Delay:** The exit door is a single point on one of the walls. Only one student can clear the door at a time, creating a small delay before the next student can exit.
- **Forward Vision only:** The students only look at others in a 90° cone in front of them.
- **Personal Space Repulsion:** All students have a small repulsive force from neighboring students within radius R_{rep}

Model Description and Equations

The model is constructed as an Agent-Based Model where the instantaneous state of each student i (for $i = 1, 2, \dots, N$) is defined by their continuous position and velocity vectors. The dynamics of the system is governed by a coupled system of first order ordinary differential equations:

$$\frac{dx_i}{dt} = v_i$$

$$\frac{dv_i}{dt} = \alpha[V_{des,i} - v_i]$$

State Variables (Updated at each time-step):

- $x_i(t) = [x_i(t), y_i(t)]^T$: Position of student i at time t
- $v_i(t) = [v_{x,i}(t), v_{y,i}(t)]^T$: Velocity of student i at time t
- $active_i(t)$: Boolean stating if student i is inside room
- $t_{available}$: Timestamp tracking when the door is unblocked

Model Constraints and Parameters:

- N : Number of students
- L : Classroom side length
- α : Sensitivity relaxation parameter dictating how quick a student accelerates
- $\Delta\tau$: Delay between students exiting
- γ : Congestion sensitivity coefficient
- v_{max} : Maximum walking velocity of a student
- R_{rep} : Radius around each student where repulsion activates
- k_{rep} : Repulsion gain constant which scales the magnitude of the pushing force
- $CONE_COS$: The cosine of 45° for defining the forward vision field

Navigation Direction

Every student continuously evaluates a normalized unit vector pointing from their current position to the door:

$$\hat{e}_i = \frac{x_{door} - x_i}{\|x_{door} - x_i\| + \epsilon}$$

We add a small nonzero term $\epsilon = 10^{-9}$ to the denominator to avoid any potential division by zero errors when the student gets very close to the door.

Visual Clearance Calculation

To determine if a student is clear to move forward towards the door, each student must see if there are others in their path. Let $v_{ji} = x_j - x_i$ be the relative displacement vector, and

$\hat{u}_{ji} = \frac{v_{ji}}{\|v_{ji}\|}$ be its normalized direction. A student j is considered an obstacle if student j satisfies

the following two conditions:

1. **Within student i 's peripheral vision:**

$$\hat{e}_i \cdot \hat{u}_{ji} \geq \text{CONE_COS}$$

This dot product tracks the cosine of the angle between the vectors. It verifies if the ratio of the adjacent side of the hypotenuse is greater than or equal to the cosine of 45°.

2. Between student i and the door:

$$||x_{door} - x_j|| \leq ||x_{door} - x_i||$$

The forward clearance ξ_i is defined as the distance to the closest obstacle minus the average human body radius (0.3):

$$\xi_i = \min(||x_j - x_i|| - 0.3)$$

If no forward obstacles are detected, ξ_i defaults to an open road distance maximum.

Optimal Velocity Function $V(\xi_i)$

The target speed magnitude is evaluated using a hyperbolic tangent relationship adapted from the Agent-Based Modeling lecture.

$$V(\xi_i) = v_{max} [\tanh(\xi_i - 0.7) + \tanh(0.7)]$$

If forward space drops to a critical state, $\Delta x_i \leq 0$, the function outputs a desired speed of 0.

Repulsion Force

To prevent students from stacking on top of each other, we must add in a repulsive vector from all students within our pre-determined social radius.

$$f_{rep,i} = \sum k_{rep} \frac{R_{rep} - d_{ij}}{d_{ij} + \epsilon} d_{ij}$$

Where $d_{ij} = ||x_i - x_j||$ is the distance between students. As $d_{ij} \rightarrow 0$, the ratio scales to infinity, creating a larger repelling force.

Total Desired Velocity ($V_{des,i}$)

Combining the navigation direction and social boundary interaction equations, the target velocity vector is:

$$V_{des,i} = V(\xi_i) \hat{e}_i + f_{rep,i}$$

Door Queueing

The door should only be able to process one student at a time, and it must require a delay for a person to physically step through the frame and clear the way. The system keeps track of a global availability timer. For any student i within the door's capture radius, their velocity is determined by the state of the door:

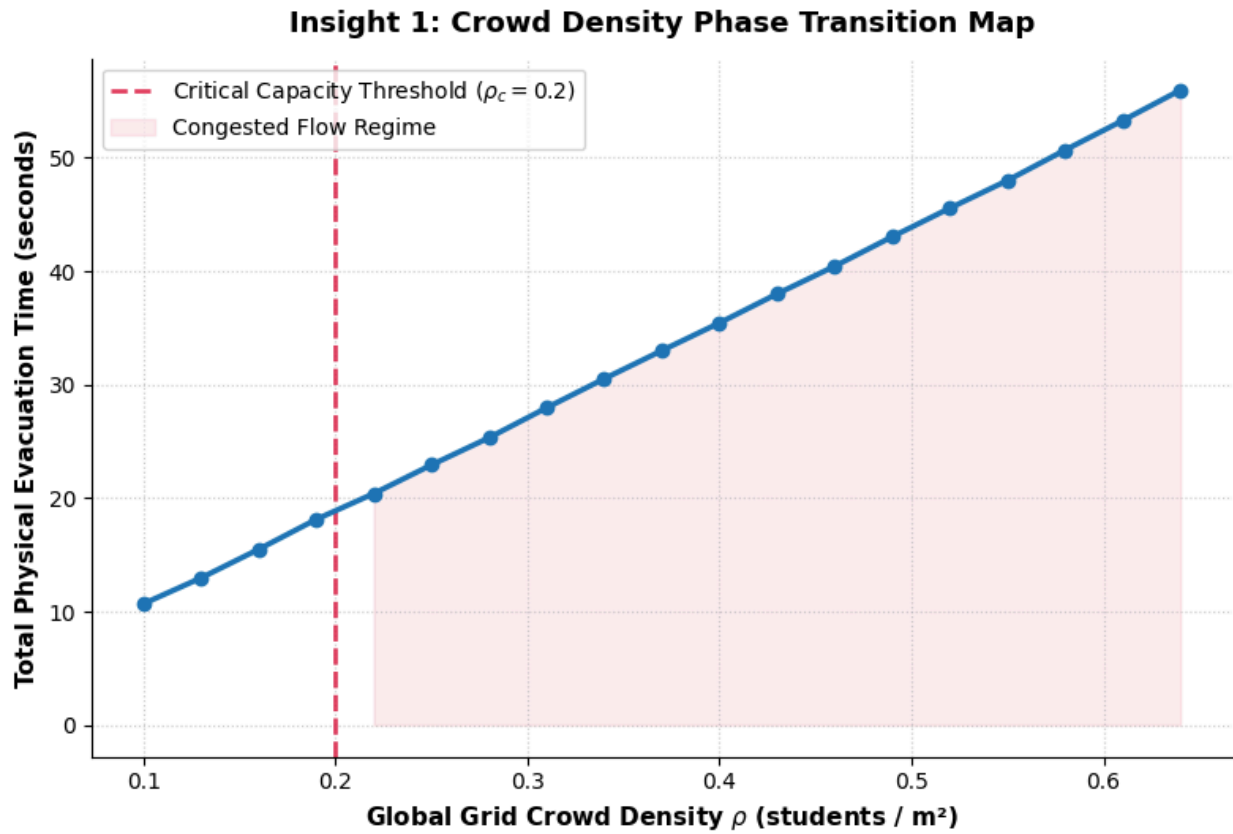
$$\begin{cases} 0, & \text{if } t < t_{\text{avail}} \quad (\text{Server Busy}) \\ V(\Delta x_i), & \text{if } t \geq t_{\text{avail}} \quad (\text{Server Vacant}) \end{cases}$$

The moment the door is vacant, the model then chooses the closest student i who minimizes the Euclidean distance to the door

$$i = \operatorname{argmin} ||x_i - x_{\text{door}}||$$

Analysis and results

Insight 1: Crowd Density Phase Transition

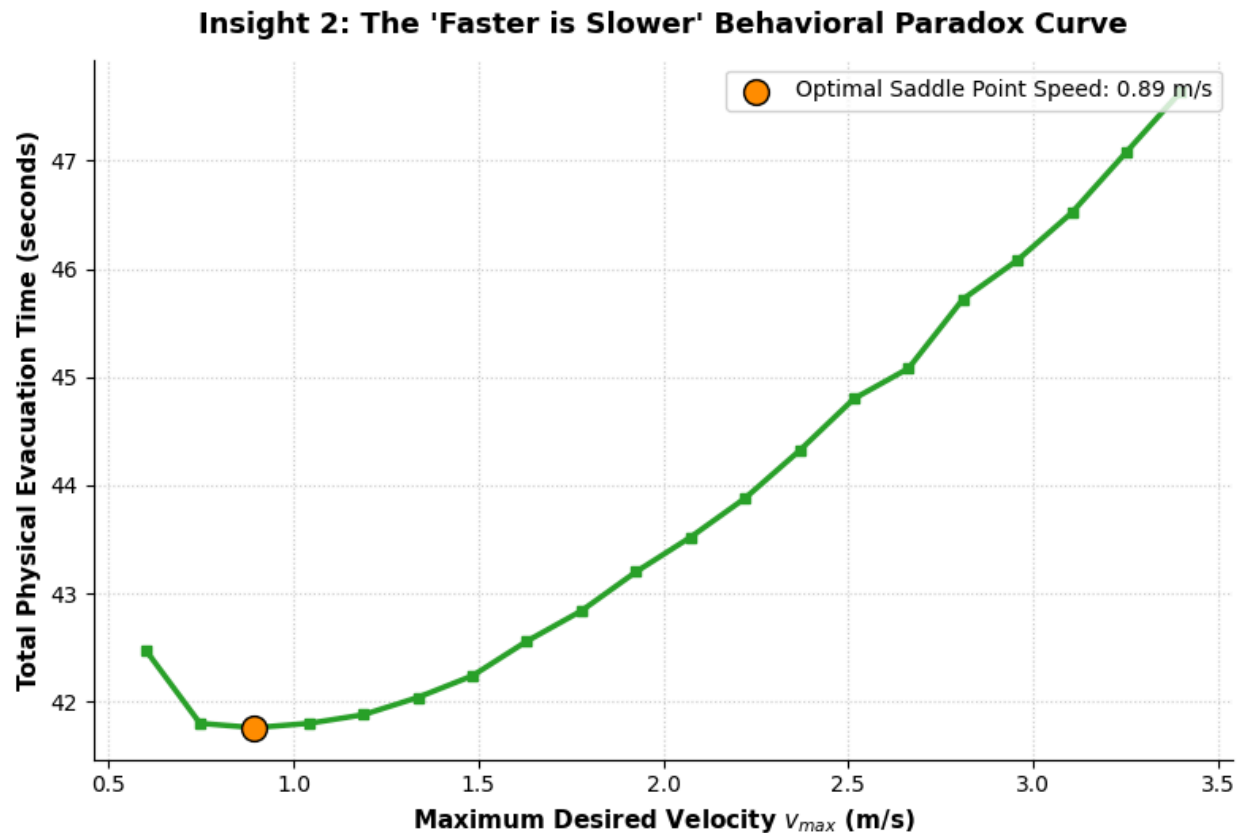


A crowd-density phase transition graph reveals that total evacuation time increases as global crowd density increases.

At low density (0.1 to 0.2 student/m²), evacuation time is relatively low with no sign of congestion. As the density moves past the critical capacity threshold $\rho_c = 0.2$ students / m², the

model suggests that the system begins to move from a free flowing evacuation state to a more congested state.

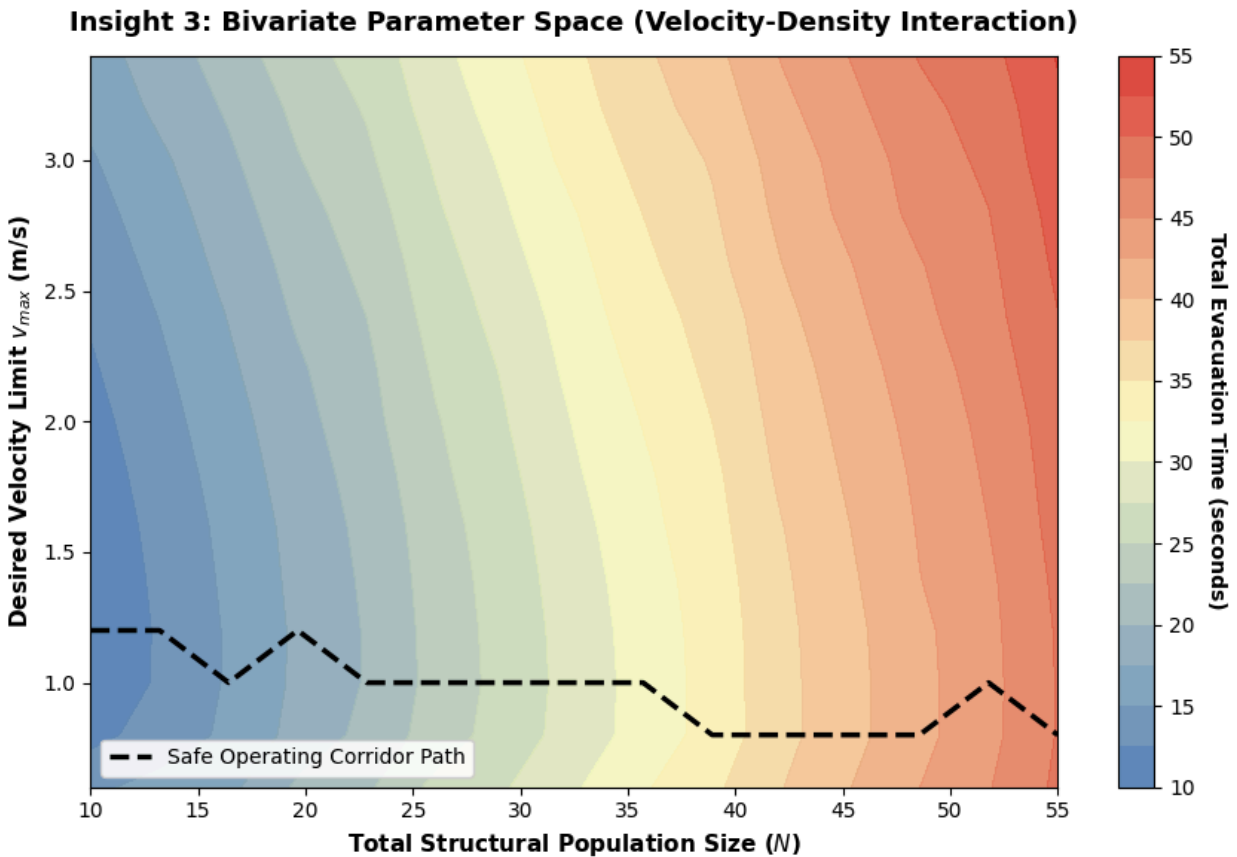
Insight 2: The "Faster is Slower" Behavioral Paradox



A sensitivity analysis on the desired walking speed v_{max} reveals a strict U-shaped optimization curve, proving human panic intuition mathematically false.

Pushing velocities beyond an optimal saddle point (~ 0.89 m/s) forces students to hit the door too aggressively. The distance between students shrinks, causing the social repulsion factor to increase. Students are pushing each other against the walls to the side of the door which is slowing the evacuation time.

Insight 3: Bivariate Phase Space (The Velocity-Density Interaction)



To understand how variables compound, we conducted a bivariate contour analysis, simultaneously sweeping both population size N and walking speed v_{max} to map the total system phase space.

The most important takeaway is that population size N is the main factor that decides how long it takes to empty the room. Because the colored bands on the graph run mostly straight up and down, you can see that simply adding more people to the room causes the evacuation time to go up in a steady, predictable line. This makes sense for a structural bottleneck: a single-door exit can only handle so many bodies per second.

However, the maximum desired velocity v_{max} introduces a hidden penalty. Instead of helping students leave faster, moving too fast actually hurts the group's total time. You can see this by looking at how the vertical color bands tilt backwards to the left near the top of the graph:

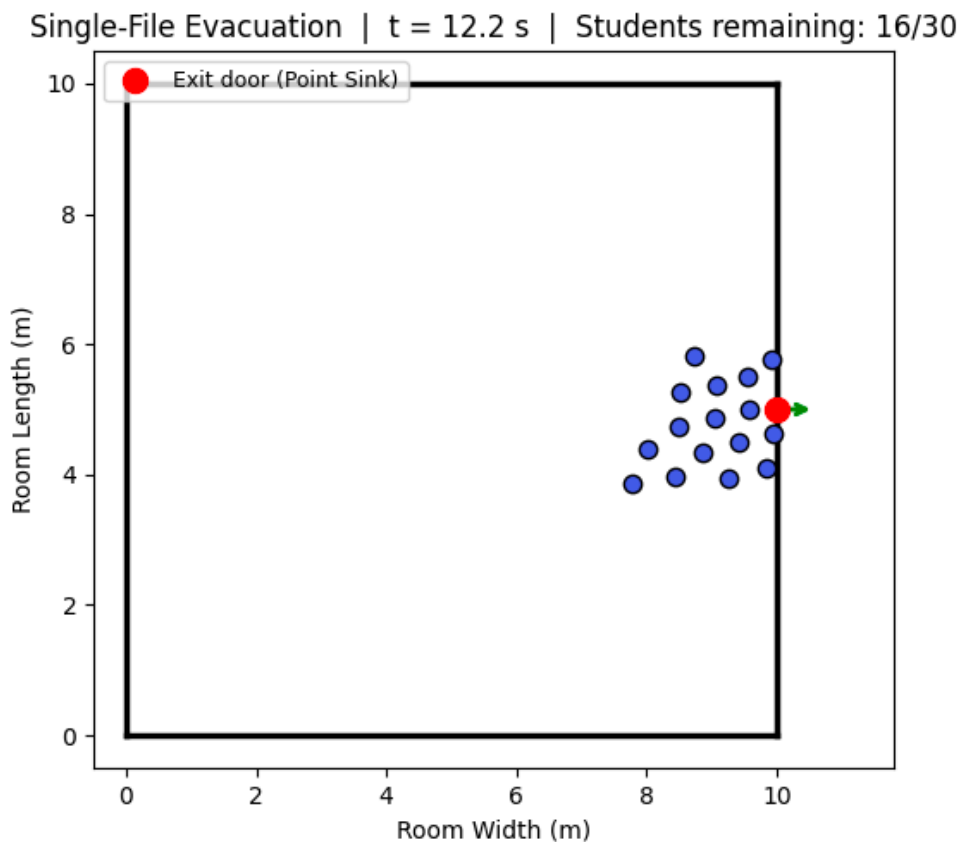
- In packed rooms $N > 40$: If you look at a room with 50 students and follow it straight up from a calm walk 1 m/s to a panicked run 3 m/s, the color on the map shifts from a safe

light orange to a dangerous dark red. The total time jumps from roughly 45 seconds to over 55 seconds.

- The Safe Operating Corridor: Because rushing makes the bottleneck worse, the black dashed line (the absolute fastest way to clear the room) stays completely stuck at the bottom of the map. It shows that the ideal speed is a calm, controlled walk between 0.8 m/s and 1.2 m/s , no matter how many people are in the room. In short, while the number of people sets the baseline layout for the traffic jam, it is the individual greed and high speed of the students that actively creates the clogging penalty. Rushing is a massive liability that slows everyone down.

Insight 4: Semi-Circle Crowd at the Door

Because our model tracks 2D coordinates, we observe the queue forming a dense semicircular arc around the door rather than a 1D line.



The highest kinetic pressure occurs exactly at the radial perimeter of the capture zone.

On the right, we have a 2D view of our model simulation that demonstrates the behavior of 30 students evacuating a classroom with sides width of 10 meters. We can see that the students form a semicircular arc in front of the door as they are queuing for their turn to leave the classroom through the single point door that only allows one student to exit every 0.8 seconds.

Interpretation and limitations

The Crowd Density Phase Transition plot demonstrates the idea that when the classroom is not very crowded, students move towards the exit and through the door with minimal interference from others, but as more and more people are in the classroom of the same size, the required time for everyone to leave increases and a crowd tends to form around the door as everyone tries to exit the room.

From the “Faster is Slower” Optimization Curve Graph illustrating the relationship between total evacuation time and maximum desired speed of individual students, the strict U-shaped curve with minimum at around 1.3 m/s for maximum desired speed suggests that faster speed does not correlate with faster evacuation time. This is because when a crowd panics and sprints, they generate immense chaotic lateral friction at the bottleneck (door). Therefore, restricting a crowd to a brisk, controlled walk mathematically minimizes total evacuation time by preserving forward momentum and preventing lateral lock-ups.

Observing the Bivariate Parameter Phase Space, the negative correlation between desired speed and population size suggests that, to minimize total evacuation time, the desired speed for each student is dependent on the number of students within the classroom. Therefore, evacuation protocols cannot be one-size-fits-all and standard operating procedures must dictate different movement profiles based on real-time room occupancy.

Finally, due to the second dimensional nature of our model, the observed semicircular arc shaped queue at the door suggests that, to prevent crushing injuries, physical obstacles (pillars, sharp desks) must be completely cleared from the immediate radial footprint of any main exit, as this is where lateral repulsion forces reach their absolute maximum.

Limitations

Some limitations to our model include the assumption that all individuals behave the same way with the same physical capabilities as they are all trying to evacuate out the same door and move at the same desired speed given the conditions for them to do so. When in reality, students can have different walking speed, reaction time, and sense of urgency to evacuate the room. Our model is also simulated on a flat, two dimensional square space with no obstacles obstructing the

paths for students to the door. In most classrooms, there are desks, chairs, sometimes uneven surfaces (stairs) and irregular wall shapes that strongly affect how students form lines, clusters, or jams.

The model assumes that the door is a single point where one student gets to leave at a time to capture the bottleneck effect. However, real doors have widths that, depending on the size of the students, can often fit through more than one student at a time. In addition, the delay between when a student is allowed to pass through the door is oversimplified, as exit delay in real life depends on several factors, including how close the students are to one another, the angle to which they exit the door, door width, etc.

The model also assumes students only respond to others in a forward vision cone and only interact through personal-space repulsion, which ignores more complex behaviors such as panic, pushing, cooperation, or group-following. Therefore, the model is useful for studying general crowding and bottleneck patterns, but its results should be interpreted as a simplified approximation rather than an exact prediction of real evacuation behavior.

Suggestions for model extensions or improvements

A possible extension of this model would be to make the student agents more heterogeneous, such as assigning different speeds, reaction time, comfort distance, and levels of urgency, instead of assuming that all students behave the same way. Another way to improve the model would be to include obstacles to represent desks, chairs, doorway, etc, to resemble more of a real life classroom setting and how they would affect the decision making of students. Also, the doorway can be given a width instead of being represented by a single point to capture realistic congestion at the door.

The model could also be extended by adding more complex behavioral rules, such as students following nearby classmates, avoiding dense crowds, turning their heads to observe more than just the area in front of them, or reacting differently under panic conditions. Finally, the parameters of the model, such as walking speed, repulsion radius, and exit delay time, could be calibrated using real evacuation data or small-scale classroom observations. These improvements would make the model more realistic and allow it to produce results that are more useful for understanding actual evacuation behavior.